

DeepArousal-Net: A Multi-block Recurrent Deep Learning Model for Proactive Forecasting of Non-Apneic Arousals from Multichannel PSG

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Abstract—Objective: This study aimed to develop a deep learning model capable of accurately forecasting non-apneic sleep arousals, which are brief awakenings that disrupt sleep continuity and contribute to daytime fatigue. **Methods:** We introduce DeepArousal-Net, a novel deep learning model designed to predict non-apnea arousals using multichannel polysomnography (PSG) records. DeepArousal-Net employs a multi-block architecture composed of convolutional neural networks (CNNs) to extract features from a comprehensive set of PSG signals, including EEG, ECG, EOG, EMG, oxygen saturation, and airflow. Bidirectional Long Short-Term Memory (Bi-LSTM) layers are incorporated to capture temporal dependencies in the extracted features. **Results:** DeepArousal-Net achieved an accuracy of 81.31%, sensitivity of 71.23%, and specificity of 81.90% in forecasting arousals 30 seconds in advance. The model demonstrated superior performance compared to traditional time-series prediction methods. **Conclusion:** DeepArousal-Net's ability to forecast non-apneic sleep arousals marks a significant advancement over existing post-event detection systems. **Significance:** By anticipating arousals, DeepArousal-Net opens new possibilities for the development of innovative interventions and personalized sleep management strategies, potentially leading to improved sleep quality and overall well-being.

Index Terms—Arousal, deep learning, convolutional neural network, forecasting, Long short-term memory, personalized sleep care

I. INTRODUCTION

SLEEP occupies approximately one-third of the human life span and is essential for health, mood, and overall well-being [1]. However, sleep problems impact almost half of the adult population and can contribute to risk for different health problems such as cardiovascular disease, mental disorders, and even cancer.

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The quality of sleep in patients with breathing-related and moving-related sleep disorders is degraded by the frequent occurrence of arousals. Arousals are temporary interruptions of wakefulness into sleep or spontaneous increase of the vigilance level during sleep [2]. According to the American Academy of Sleep Medicine (AASM) guideline, electroencephalographic arousals are defined as abrupt shifts of electroencephalography (EEG) frequency, including alpha, theta, and/or frequencies greater than 16 Hz that last at least 3 seconds, with at least 10 seconds of previous stable sleep [3].

Arousals are often the result of obstructive sleep apnea (OSA) or hypopnea events or are respiratory effort-related (RERA). A number of other pathological arousals have also been documented including teeth grinding (bruxism), body movements, and pain. RERA is the most common cause of non-apnea and non-hypopnea arousals [4]. Arousals disrupt sleep architecture, affecting hormonal balance, glucose metabolism, blood pressure regulation, and stress levels [5], [6]. This disruption is linked to various health issues, including cardiovascular problems, metabolic disorders like diabetes, mental health conditions such as anxiety and depression, obesity, and hypertension [7].

Polysomnography (PSG) is a widely used method for monitoring and diagnosing sleep disorders in sleep laboratories. PSG measures multiple physiological signals, such as electroencephalogram (EEG), electromyogram (EMG), electrooculogram (EOG), electrocardiogram (ECG), blood oxygen saturation level (SaO₂), respiratory airflow (airflow), and respiratory movement (chest ABD), to investigate sleep architecture and detect respiratory or non-respiratory-related sleep disorders [8]. While PSG effectively detects sleep disorders, recent advancements in wearable sensors, flexible electronics, and "big data" predictive analytics have advanced the point-of-care (POC) technologies in the treatment of different sleep disorders such as OSA and RERA arousals [9]. These include multiple variants of automatic positive airway pressure (APAP) machines, position-adjusting beds, and nerve stimulation devices [10]–[12]. Such devices employ biomedical sensors to gather signals and extract certain signal patterns using intelligent algorithms to detect or predict the occurrence of different sleep events and conditions. Appropriate actions or interventions are then taken to alleviate the effect of sleep disorders and improve sleep quality. Within this context, being able to predict the occurrence of sleep events ahead of time provides the opportunity to take appropriate actions in time

to manage, control, and treat them [4], [13]. For example, early detection of abnormal breathing can help APAP devices in providing the appropriate pressure level more effectively [14]. This proactive approach, made possible by accurately predicting sleep events, represents a significant shift from reactive to preventive sleep care. However, to date, the focus has been on the detection of sleep arousals and forecasting other sleep events such as OSA, highlighting a critical gap in the field regarding forecasting arousals.

In this paper, we introduce DeepArousal-Net, an innovative deep neural network-based method for proactively forecasting sleep arousals. DeepArousal-Net employs a unique multi-block architecture composed of convolutional neural networks (CNNs) to extract informative features from a comprehensive set of physiological signals derived from multichannel polysomnography (PSG), including EEG, EOG, EMG, airflow, and SaO₂. Each CNN block incorporates convolutional layers with varying kernel sizes, max-pooling layers to downsample feature maps, and dropout layers to prevent overfitting. Within each block, bidirectional Long Short-Term Memory (Bi-LSTM) units are integrated to capture temporal dependencies among the extracted features, enabling the model to learn intricate patterns that precede arousal events. These features are then synthesized through a fully-connected neural network, culminating in a softmax layer for predicting sleep arousal episodes. To evaluate our proposed method, we leveraged a dataset comprising 994 PSG recordings from the Sleep Lab of Massachusetts General Hospital, a rich resource provided specifically for the Physionet/CinC 2018 challenge. Adhering to the guidelines of this competition, our research focused on forecasting RERA-related and non-RERA, non-apnea arousals. We evaluated the performance of our forecasting model using key metrics, including accuracy, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), area under the receiver operating characteristic curve (AUROC), and area under the precision-recall curve (AUPRC).

The organization of the remainder of the paper is as follows. In Section II, the related research is reviewed. The dataset and its characteristics, methods used for preprocessing of PSG signals, and the algorithm proposed to forecast the occurrence of arousals are discussed in Section III. Section IV presents the experimental results obtained on our dataset, and section VI concludes the paper.

II. PREVIOUS WORK

The need for efficient, fast, and reliable algorithms for detecting arousals has led to the development of several automatic detection methods that process various physiological signals [15]–[19]. De Carli et al. introduced an automatic arousal detection method using two EEG channels and an EMG signal, applying wavelet transform across six frequency bands [15]. This method outperformed human experts in sensitivity (88.1% vs. 72.4% and 78.4%) by identifying more arousals, but had lower selectivity (74.5% vs. experts' 83.0% and 82.0%). The comparison involved a two-stage validation process where EEG experts independently and then jointly reviewed the recordings to establish a reference set of arousals,

highlighting the system's potential when combined with expert validation for enhanced accuracy. Popovic et al. presented an algorithm for detecting arousals using a single-channel frontal EEG recorded from 29 healthy adults [16]. Their approach involved spectral decomposition of EEG signals using a decision tree, computation of descriptors of sleep macro and microstructure, artifact detection, and classification of 30-second epochs into sleep stages. The accuracy of this approach was above 80% when compared with four different expert scores. However, their analysis was limited to only 29 individuals. Shmiel et al. proposed a method for detecting arousals based on sequential pattern discovery using EEG, EMG, and SaO₂ signals [17]. Their approach achieved a sensitivity of 75% and a precision of 77% when tested on a dataset of PSG signals obtained from 20 individuals. Zahid et al. proposed MSED, a joint multimodal deep learning model capable of detecting arousals, leg movements, and sleep-disordered breathing from PSG recordings, achieving an F1 score of 0.70 for arousals while significantly reducing model size [20]. Li et al. developed a deep learning framework using only ECG data to detect cortical arousals, achieving an AUPRC of 0.62 and demonstrating its feasibility for home-based sleep testing [21]. Similarly, Mishra et al. introduced a 1D CNN-based U-Net architecture that captured long-range temporal dependencies in PSG signals, reporting strong results with an AUPRC of 0.49 and AUROC of 0.94 [22]. Another study by Li et al. presented CSleep, a multitask model using only PPG signals and cross-granular attention to jointly recognize sleep stages and arousals, achieving an AUPRC of 0.63 on the MESA dataset [23].

While these studies have made significant contributions to arousal detection, the majority of research has focused on analyzing pre-recorded physiological signals. There is a lack of research on forecasting future sleep arousals, which could have a major impact on the development of preventive strategies for sleep disorders. Predicting arousals before they occur could enable proactive interventions, shifting the focus from reaction to prevention. For example, forecasting arousals could be crucial in optimizing the performance of CPAP machines for individuals with OSA, developing adaptive nerve stimulation techniques to prevent arousals triggered by external factors, and personalizing sleep management strategies by identifying and mitigating individual risk factors.

Despite the potential benefits, research on forecasting sleep events has been relatively limited, primarily focusing on predicting apneic events [24]–[28]. For instance, Le et al. [24] proposed a nonparametric statistical Dirichlet-process mixture-Gaussian-process model for predicting the onset of OSA. The model achieved accuracies ranging from 67% to 80% in predicting apnea episodes 1 to 5 minutes before their onset. Dagum et al. [25] employed dynamic network models to forecast apneic events using a Bayesian belief network, analyzing the multivariate time series of chest volume, heart rate, and oxygen saturation data. Bahrami and Forouzanfar [28] proposed a novel framework for predicting the occurrence of apnea from a single-lead ECG signal using deep recurrent neural networks (RNNs). The proposed approach involved the extraction of ECG R-peak amplitudes and R-R intervals using

power spectral analysis. These features were then aligned and used to develop deep RNN models that can extract the most predictive ECG features and forecast the occurrence of apneic events.

Given the lack of research in the area of forecasting sleep events, particularly arousals, there is a significant need for the development of intelligent algorithms. These algorithms should be capable of predicting the occurrence of these events in the future based on the patterns observed in nocturnal physiological signals.

III. MATERIALS AND METHODS

A. Dataset

The dataset used in this study was sourced from the Sleep Lab of Massachusetts General Hospital, through the Computational Clinical Neurophysiology Laboratory and the Clinical Data Animation Center. This dataset was made available for the Physionet/CinC challenge in 2018 [29], [30] and consists of 994 PSG recordings of individuals. The participants' age varies from 18-90 years, with an average of 55 years. 67% of the individuals are male (666), and 33% are female (328). The average body mass index (BMI) is 33 kg.m-2. More details on the characteristics of participants in the dataset are summarized in Table I and Table II [29].

The dataset consists of thirteen different physiological signals including 6-channel EEG (F3-M2, F4-M1, C3-M2, C4-M1, O1-M2, and O2-M1), left eye EOG, EMG obtained at the chin, respiratory movements at the chest and abdomen, ECG, SaO2, and airflow (see Fig. 1).

The dataset underwent annotation by a group of seven scorers, with each PSG assigned to a single scorer. The scorers marked the beginning and end of arousals, along with their corresponding types as per the table. Moreover, the dataset also underwent annotation for wakefulness (W), and the four sleep stages, namely rapid eye movement (REM) sleep, non-REM stage 1 (N1), non-REM stage 2 (N2), and non-REM stage 3 (N3).

Importantly, the PhysioNet/CinC Challenge guidelines explicitly focused on forecasting arousals not associated with apneas or hypopneas. Thus, this study adheres to the challenge's criteria by targeting non-apneic arousals only, including spontaneous arousals and respiratory effort-related arousals (RERAs). Events directly related to obstructive, central, or mixed apneas were excluded from the target label set. This design choice not only aligns with the challenge objectives but also facilitates direct performance comparison with state-of-the-art (SOTA) methods developed using the same dataset and task definition. It also allows us to focus on a clinically underexplored class of arousals that occur independently of respiratory events, which may benefit from alternative detection and intervention strategies beyond CPAP therapy.

The criteria for annotating arousal regions (target arousal) adhered to the following guidelines [29]:

- For a RERA-related arousal: from 2 seconds before the arousal begins to 10 seconds after it ends.
- For non-RERA and non-apnea related arousals: from 2 seconds before the arousal begins to 2 seconds after it ends.

TABLE I
CLINICAL CHARACTERISTICS OF THE DATASET

Clinical Feature	Average
Sample size	994
Age	55(14.3)
Sex (% male)	67
BMI	33 (7.8)
AHI	19 (14.6)
ESS	8.5 (5.3)
Recording time (h)	7.7 (0.7)
Time in bed (h)	7.5 (0.7)
Sleep time (h)	6.2 (1.1)
Drug Usage	%
Hypertension	41.0
Sleep aids	29.0
Antidepressant	25.7
Neuroactive	20.8
Benzodiazepine	16.9
Diabetic	11.9
Opiate	8.1
Antihistamine	4.8
Stimulant	3.9
Neuroleptic	4.5
Herbal	4.3
Reason for visit	%
Diagnostic	41.16
Split night CPAP	37.95
All night CPAP	20.88

^a BMI: Body Mass Index, Kg.m-2. ^b AHI: Apnea-Hypopnea Index, events/hr. ^c ESS: Epworth Sleepiness Scale.

^d CPAP: Continuous Positive Airway Pressure.

An illustration of a 30-second PSG recording with sleep arousal annotation can be observed in Fig. 2.

B. Preprocessing

This study used a data set consisting of five selected PSG channels to capture various physiological signals from patients. These channels include the EEG, which measures the electrical activity of the brain, the EOG, which measures the resting electrical potential between the cornea and Bruch's membrane, the EMG, which measures muscle response or electrical activity in response to a nerve stimulation of the muscle, the SaO2, which measures the percentage of hemoglobin molecules in the arterial blood that are saturated with oxygen, and the transitional Air Flow, which measures the rate of airflow during respiration.

According to the AASM manual for the scoring of sleep and associated events, the primary EEG derivatives used to monitor the electrical activity of the frontal, central, and occipital regions of the brain are F4-M1, C4-M1, and O2-M1 [3]. It is worth noting that PSG recordings can be configured to display either a montage containing all three EEG derivations (M3EEG) - F4/M1, C4/M1, and O2/M1 - or a montage containing only one EEG derivation (M1EEG) at C4/M1, as mentioned in [33]. For the purposes of our study, we specifically chose the C4/M1 channel for predicting the occurrence of arousal events.

TABLE II
SLEEP AND AROUSAL CHARACTERISTICS OF THE DATASET

Time spent in sleep stage	%
Wake	28.0
NREM 1	19.6
NREM 2	51.0
NREM 3	14.0
REM	15.5
Number of target arousals	count
Bruxism	30
Cheyne-Stokes breathing	3
Hypoventilation	4
Noise	1
Partial airway obstruction	11
PLM	36
RERA	43,822
Snoring	28
Spontaneous	70
Number of non-target arousals	count
Hypopnea	56,936
Central apnea	22,763
Mixed apnea	2,641
Obstructive apnea	32,547

^a PLM: Periodic Leg Movement,

^b RERA: Respiratory Effort-Related Arousals

To ensure the quality of our PSG data, we meticulously preprocessed all signals, with a particular focus on the EEG:

1) *Filtering and Resampling*: To prepare the physiological signals for analysis, a series of preprocessing steps were applied to enhance signal quality and optimize computational efficiency. First, to prevent aliasing during subsequent down-sampling and to eliminate unwanted noise, a 6th-order infinite impulse response (IIR) high-pass filter with a cutoff frequency range of 25 Hz was applied. This cutoff was specifically chosen to retain the relevant information while ensuring that frequencies above the Nyquist limit of the target sampling rate were attenuated. Following filtering, all signals, initially sampled at a frequency of 200 Hz, were down-sampled to a frequency of 50 Hz.

2) *Outlier Removal*: To ensure model stability, especially with our limited dataset, and to mitigate the impact of severe movement artifacts, a targeted outlier removal procedure was applied to the EEG signals.

EEG signals were segmented into 8-second epochs. Epochs where the absolute amplitude exceeded the 95th percentile were excluded, as these were highly indicative of gross movement artifacts [34], [35]. While deep learning models can learn to handle some noise, these extreme outliers can introduce high-variance disruptions that particularly challenge learning when forecasting rare events in smaller datasets. When an EEG segment was removed, corresponding 8-second segments across all other PSG channels were also excluded to maintain data integrity.

3) *Standardization*: To ensure that the features extracted from each channel were comparable and equally significant, we standardized the data by subtracting the mean and dividing by the standard deviation, resulting in a mean of zero and a

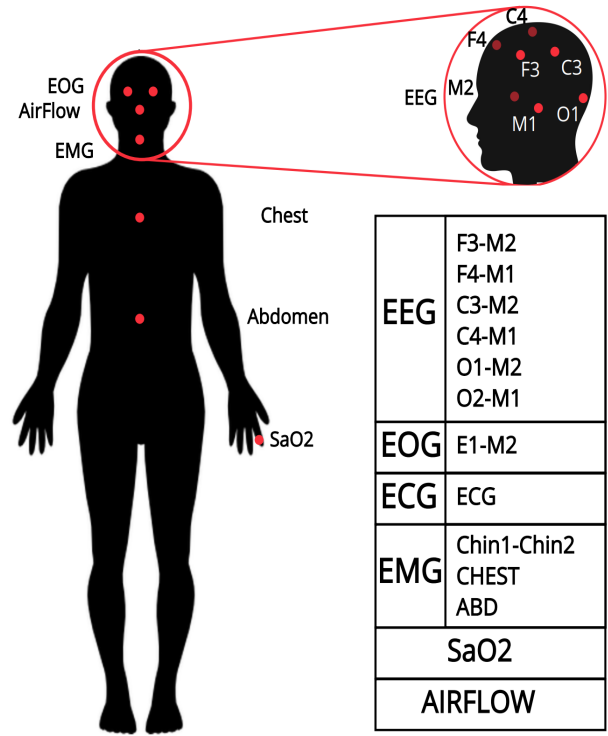


Fig. 1. PSG electrode placement. The location of each 13-channel polysomnogram is shown, including electrode placement scheme on the scalp, known as International 10–20 system, that was recommended by the International Federation of Societies for EEG and Clinical Neurophysiology (IFSECN) [31], [32] (Electrodes that are used in the dataset are shown)

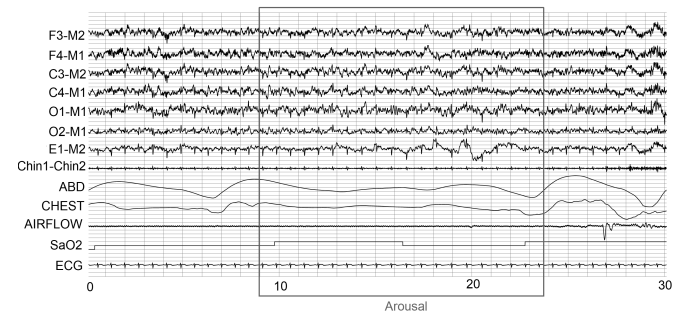


Fig. 2. Example of a PSG recording during 30 seconds of sleep. The square represents the onset of arousal according to the annotation provided by the PhysioNet/CinC Challenge 2018.

standard deviation of one across all channels. This preprocessing step is crucial for achieving optimal performance and accuracy in our analysis.

C. Data Preparation for Deep Learning

To label the signals, we partitioned each signal into 30-second time intervals, as illustrated in Fig. 3, and assigned each interval a binary label of arousal or non-arousal. As sleep arousal events are infrequent, comprising only 10% of the entire dataset [36], maintaining balance in the dataset is critical. To address this issue, we extracted an equal number of non-arousal segments as arousal ones, ensuring that half

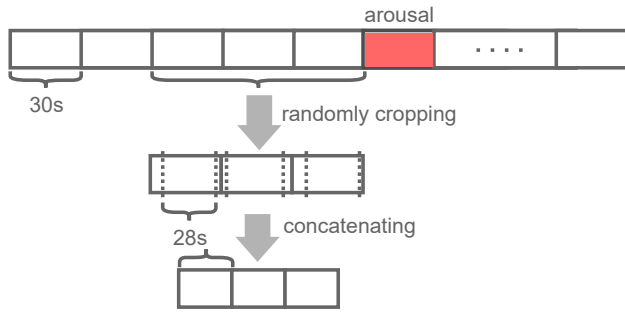


Fig. 3. Data preparation process. The raw signal is analyzed in 30-second intervals. To ensure that the model is not biased towards any specific portion of the input data and avoid overfitting, every 30 seconds of data is randomly cropped into 28-second windows, which are then concatenated to form the final input data.

of the batch size consisted of windows marked as arousal, and the other half was comprised of the remaining windows. By using this approach, we trained the neural network on a balanced dataset, which is crucial for obtaining accurate and reliable results.

We employed various combinations of data lengths from the past (90 sec, 180 sec, and 270 sec) and forecasting horizons (30 sec, 60 sec, and 120 sec) for our predictive modeling. This allowed us to evaluate the model's performance across a range of time horizons and determine the optimal input data length and forecasting horizon for the given task.

To enhance the generalizability of our model and mitigate overfitting, we employed a data augmentation technique based on random cropping. Within each 30-second interval, we randomly selected a 28-second segment using a random offset (see Fig. 3). This cropping technique introduces controlled variability into the training data, preventing the model from over-relying on specific portions of the input and thus promoting greater robustness.

The cropped segments were then concatenated to form the input data for our deep learning model. By incorporating this technique into our data preparation pipeline, we are able to improve the model's performance and ensure its applicability to a wider range of datasets.

D. DeepArousal-Net

We introduce DeepArousal-Net, a novel deep learning architecture featuring a multi-block design to process PSG signals from different channels and proactively forecast sleep arousals, as illustrated in Fig. 4. Our model utilizes CNNs, which have demonstrated exceptional proficiency in processing multichannel biomedical signals through a series of filters, generating feature maps at each layer, and enabling automated feature extraction without the need for manual feature engineering. Additionally, we integrate RNNs into our methodology to capture the temporal patterns of physiological signals acquired during sleep.

To process the preprocessed multichannel physiological signals, we utilized five distinct CNNs, with each responsible for processing one input channel. Each CNN, consisted of six

TABLE III
CNN ARCHITECTURE

Conv1D layer	Number of Kernels	Kernel Size	Max-Pooling
Block 1	8	16	NO
Block 2	8	16	YES
Block 3	8	16	NO
Block 4	16	32	YES
Block 5	32	32	NO
Block 6	32	64	YES

sub-modules comprising two architectures, namely M1 and M2 (see Fig. 4). M1 module included a 1D convolutional layer with parametric rectified linear unit (PReLU) activation and dropout (DO). M2 module included a 1D convolutional layer with max pooling (MP), PReLU activation and DO. Each CNN sub-module consisted of several layers, including convolutional layers with different numbers of kernel and kernel sizes, as listed in Table III. The PReLU activation and DO layers were mainly used to enhance the efficiency of the network and avoid overfitting. The MP layer, a crucial component of our CNN architecture, played an essential role in reducing the dimensions of the feature maps generated by the convolutional layer. By selecting the maximum value from each region of the feature map covered by the filter, the MP layer highlighted the most prominent features of the previous layer's output. This process reduced the number of parameters that needed to be learned and the computation required for subsequent layers.

Following the CNNs, we incorporated bidirectional long short-term memory (Bi-LSTM) layers, dense layers, PReLU activation layers, and DO layers into each block (M3 modules in Fig. 4). These layers worked together seamlessly to extract relevant features from each channel and embed them into a lower-dimensional feature space.

We opted to use Bi-LSTM networks because of their unique ability to maintain a cell state that retains information about both long-term and short-term changes in the input features. This feature allows the network to incorporate temporal context, such as prior sleep stages or frequency shifts, which makes it particularly effective at detecting arousals, as noted in [36]. By leveraging the power of Bi-LSTMs, our model is able to capture the intricate temporal dynamics of the data and make accurate predictions. Each Bi-LSTM was comprised of two LSTM layers: one for iterating over the input in the forward direction, and another for iterating over them in the backward direction. In our model, we utilized a Bi-LSTM layer for each channel. Specifically, each channel consisted of 175 time sequences with dimensions 32, and the output dimensions were set to 256.

Following the application of CNN and Bi-LSTM modules, in each processing block, we employed dense layers with PReLU activation functions and incorporated DO layers with a rate of 0.33 for each channel. Specifically, for all channels, we constructed a network with three fully-connected layers consisting of 256, 128, and 128 units, respectively.

After applying the convolutional and LSTM layers, the

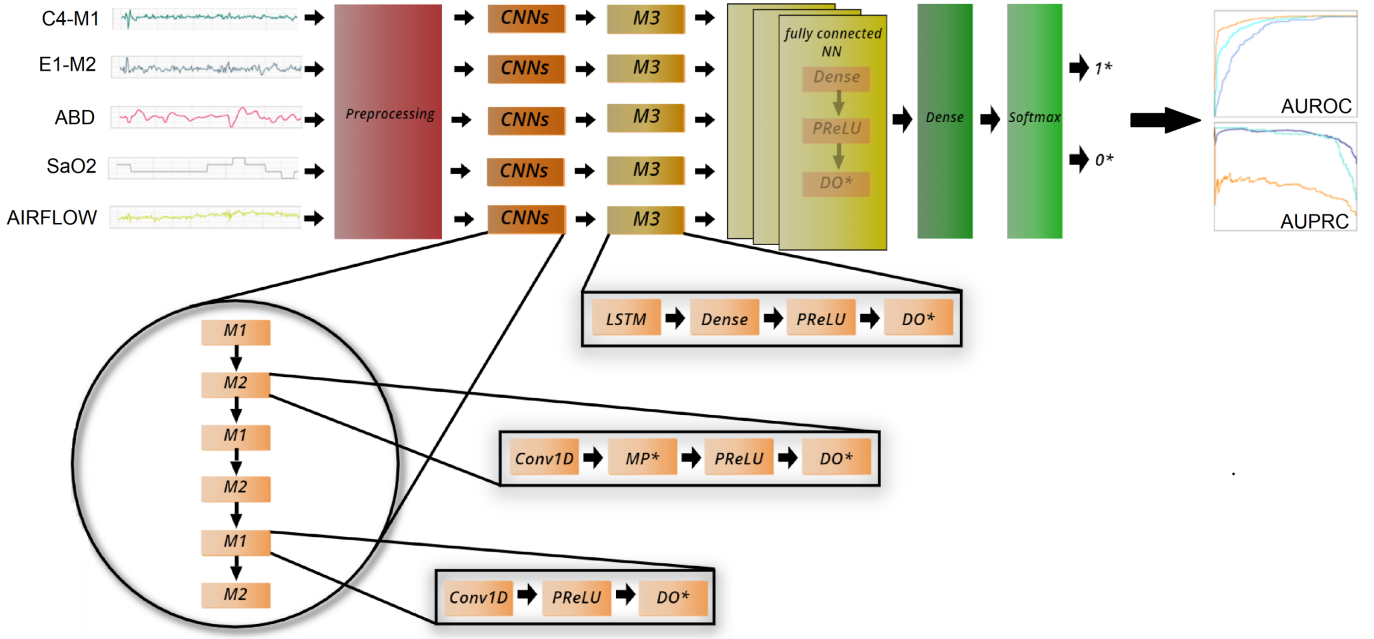


Fig. 4. DeepArousal-Net Architecture for Forecasting Sleep Arousals: This flowchart illustrates the multi-block CNN-LSTM model, which extracts features from patients' multichannel PSG data (including EEG, EOG, EMG, airflow, and SaO2) using convolutional and Bi-LSTM layers. Each CNN block (M1 and M2) consists of convolutional layers with varying kernel sizes, max-pooling (MP) layers, dropout (DO) layers, and Parametric Rectified Linear Unit (PReLU) activation functions. The extracted features are then processed by Bi-LSTM layers, followed by dense layers with PReLU activation and DO. These features are then synthesized through a fully-connected neural network, culminating in a softmax layer for predicting arousal (=1) and non-arousal (=0) events.

resulting feature vectors were combined using a non-linear approach, that included three fully connected neural networks (NNs). Each NN consisted of Dense layers with 1024, 512, and 256 units, respectively, with PReLU activation functions and incorporating DO layers for enhanced performance. The Dense layer was a standard fully connected layer that applied a linear transformation to the input data, while the PReLU activation function added non-linearity to the output improving the expressivity of the network. The DO layer was used to prevent overfitting by randomly dropping out a fraction of the input units during training. By combining these layers, the neural networks were able to learn complex representations of the concatenated feature vector. These neural networks worked together to learn high-level representations of the data.

Following the merge of the feature vectors with fully connected neural networks, the final prediction was obtained using a 2 unit Dense layer and a softmax activation function. This function converted the output of the neural network into a probability distribution over the possible classes, allowing us to predict the most likely class for a given input.

We used cross-entropy loss as the cost function to measure the difference between the true labels and predicted labels.

The complete DeepArousal-Net model contains approximately 3.2 million trainable parameters. To support reproducibility, the full implementation of DeepArousal-Net, along with training scripts and preprocessing code, is available at: https://github.com/zahrayousefijamarani/arousal_detection.

E. Experimental setup

We developed the neural network model using the Keras library [37] in Python, with a Tensorflow [38] backend. The model was trained on a Windows 10 PC with an AMD Ryzen™ 7 5800X CPU @3.8 GHz, NVIDIA 3080 GPU, and 32 GB RAM.

In order to ensure the reliability and accuracy of our findings, we implemented a rigorous 10-fold cross-validation methodology. This approach involved partitioning our dataset into 10 equally-sized folds. During each iteration, 80% of the data was allocated for training the model, while 10% was used as a validation set for hyperparameter tuning and to mitigate the risk of overfitting. The remaining 10% was reserved for testing the model's performance. The averaged results obtained from the test sets were then reported.

Furthermore, we explored the influence of different data lengths from both past and future time periods. For the forecasting task, we assessed the model's performance using past data lengths of 90 seconds, 180 seconds, and 270 seconds, with a specific focus on predicting events occurring within the subsequent 30 seconds. Additionally, we investigated future data lengths of 30 seconds, 60 seconds, and 120 seconds, taking into consideration the preceding 180 seconds. This comprehensive evaluation allowed us to analyze the model's predictive capabilities across various time frames and determine the optimal data lengths for accurate forecasting.

Our approach involved a systematic fine-tuning of several hyperparameters to achieve optimal performance in our study. We focused on two key aspects: the configuration of the convolutional layers and the fully connected neural networks.

For the convolutional layers, we experimented with different numbers of kernels and kernel sizes. The details of the hyperparameters we used are listed in Table III. By iteratively adjusting these hyperparameters, we aimed to find the combination that yielded the best results.

In the fully connected neural networks, we specifically investigated the number of layers and the number of dense units within each layer. After conducting a series of experiments, we found that a three-layer architecture with 1024, 512, and 256 units respectively produced the most favorable outcomes.

Regarding the M3 block shown in Fig 4, we also optimized the hyperparameters for the dense layers. We determined that employing three dense layers with 256, 128, and 128 units respectively resulted in the highest performance. To ensure consistency and reproducibility, we evaluated multiple fixed training epochs (40, 50, 60, 70) and found 50 to offer the best balance between convergence and generalization.

During the training process, we used the Adam optimizer [39] with an initial learning rate of $5e-5$. Adam is a popular optimization algorithm that computes adaptive learning rates for each parameter based on the first and second moments of the gradients. This adaptive learning rate scheme allowed the optimizer to converge faster and more reliably than traditional gradient descent methods. By tuning the learning rate based on the validation set, we were able to find a balance between achieving rapid convergence and avoiding the issue of the model parameters oscillating around the optimal solution.

F. Evaluation Metrics

In this work, we use accuracy, sensitivity, specificity, PPV, NPV, AUROC, and AUPRC as the evaluation metrics.

Accuracy is the overall proportion of correct predictions given by: $Accuracy = (TP + TN) / (TP + TN + FP + FN)$, where TP (true positive) is the number of correctly classified positive cases, TN (true negative) is the number of correctly classified negative cases, FP (false positive) is the number of negative cases that were incorrectly classified as positive, and FN (false negative) is the number of positive cases that were incorrectly classified as negative.

Sensitivity (recall) is the proportion of true positive cases that are correctly identified as positive, while specificity is the proportion of true negative cases that are correctly identified as negative, given by: $Sensitivity = TP / (TP + FN)$, $Specificity = TN / (TN + FP)$

PPV is the proportion of positive cases that are truly positive, while NPV is the proportion of negative cases that are truly negative, given by: $PPV = TP / (TP + FP)$, $NPV = TN / (TN + FN)$

AUROC is a performance metric used to evaluate classification models by showing the trade-off between true positive rate and false positive rate, given by: $AUROC = \int_0^1 TPR(FPR^{-1}(t))dt$, where TPR (true positive rate) is the proportion of true positive cases that are correctly identified as positive, and FPR (false positive rate) is the proportion of negative cases that are incorrectly classified as positive. "t" represents a threshold value used for determining the TPR and FPR that represents the probability threshold used to classify a prediction as positive or negative.

AUPRC is useful for imbalanced data, as it shows the trade-off between precision and recall, given by: $AUPRC = \int_0^1 P(R^{-1}(t))dt$, where P is precision or PPV and R is recall or sensitivity.

IV. RESULTS

To evaluate the effectiveness of our system, we tested the algorithm on sleep data collected from 994 individuals (described in Section III-A) using a 10-fold cross-validation technique (described in Section III-E). We evaluated the performance of our sleep arousal forecasting model in terms of its ability to predict arousal events across various time intervals in the future, while considering different lengths of past data.

A comparative analysis of our forecasting model's results is presented in Table IV, specifically examining the impact of using different time intervals in the past on the AUPRC, AUROC, accuracy, sensitivity, specificity, PPV, and NPV. By comparing the results across different time intervals, we gain valuable insights into the optimal time window that provides the most accurate forecasting results. This analysis enables us to identify the specific time frames that significantly impact the accuracy of our predictions, helping us to fine-tune our model to improve its performance.

The best performance was achieved using data from 180 seconds prior to the arousal event, out of the three time intervals tested with accuracy of 81.31%, Sensitivity of 71.23%, specificity of 81.90%, PPV of 18.91%, NPV of 97.96%, AUROC of 0.88, and AUPRC of 0.39.

A. Impact of Analysis Interval

Shortening the data length from 180 seconds to 90 seconds caused a 15.98% decrease in accuracy, 24.50% decrease in sensitivity, 15.41% decrease in specificity, 49.57% decrease in PPV, 1.87% decrease in NPV, 11.36% decrease in AUROC, and 20.51% decrease in AUPRC. On the other hand, increasing the data length beyond 180 seconds caused a 17.07% decrease in accuracy, 11.60% decrease in sensitivity, 17.35% decrease in specificity, 45.90% decrease in PPV, 1.09% decrease in NPV, 15.90% decrease in AUROC, and 17.95% decrease in AUPRC.

These results suggest that the proposed method is most effective when using a past data length of 180 seconds. Longer time intervals may contain more information about the arousal occurrence through the pattern of sleep physiological data, but their use may also increase network input data dimension and complexity, resulting in decreased network performance.

For PPV, a past data length of 180 seconds resulted in the highest value of 18.91%, compared to 9.53% and 10.23% for past data lengths of 90s and 270s, respectively. On the other hand, for NPV, a past data length of 180 seconds resulted in the highest value of 97.96%, compared to 96.13% and 96.89% for past data lengths of 90s and 270s, respectively. The highest PPV was achieved using 180 seconds of past data, indicating that this window length is most effective at identifying true arousal events. However, the NPV was high across all time intervals tested, suggesting that our method is generally effective at ruling out false alarms.

B. Impact of Forecasting Time Interval

We also examined our algorithm using various forecasting time intervals in the future. Table V provides a comprehensive summary of our results. It shows that the best performance is achieved with a 30s prediction horizon. The accuracy of the model for the 30s horizon was 81.31%, which is substantially higher than the other two horizons. The sensitivity and specificity of the model for the 30s horizon were 71.23% and 81.90%, respectively, which is the highest among the three horizons. Moreover, the AUROC for the 30s horizon was 0.88, which is relatively higher than the other two horizons. Similarly, the AUPRC for the 30s horizon was 0.39, which is higher than the other two horizons.

The table also shows that as the prediction horizon increases, the performance of the model decreases. For instance, the accuracy of the model for the 60s and 120s horizons was 64.01% and 62.23%, respectively, which is significantly lower than the accuracy for the 30s horizon. Similarly, the sensitivity and specificity of the model for the 60s and 120s horizons were lower than those for the 30s horizon.

The decrease in performance as the prediction horizon increases is likely due to the fact that the model has less data to work with as the time interval between the last data point and the arousal event increases. This suggests that the proposed method is most effective when used to predict arousal events with short prediction horizons.

Based on the data presented in Tables IV and V, our algorithm's performance is strongest when using data from 180 seconds prior to the event and predicting arousal occurrence in the next 30 seconds. This particular combination of time sections produces notably accurate prediction results, as demonstrated by the metrics presented in both tables. Specifically, we observe high AUPR and AUROC, accuracy, sensitivity, and specificity, among other key performance measures. Thus, we can conclude that our algorithm is particularly effective when used with these specific time intervals, making it a promising tool for early detection and management of sleep-related disorders and in particular sleep arousals.

C. Ablation Studies

An ablation study was conducted to evaluate the impact of our data cropping augmentation technique. The results indicated that both accuracy and AUPRC decreased by approximately 4.2% and 7.5%, respectively, when the augmentation was removed. These findings highlight the important role of augmentation in improving the model's generalizability and robustness, particularly in the context of limited subject diversity. In future work, we plan to explore more advanced augmentation strategies, such as time-warping, signal mixing (e.g., Mixup), and synthetic perturbations tailored to physiological signal characteristics.

An experiment was also conducted where the Bi-LSTM block was removed from the DeepArousal-Net architecture. In essence, removing the Bi-LSTM would transform the architecture from a sequential processing pipeline (CNN → Sequence Aggregation by LSTM → FC) to a feature extraction and direct classification pipeline (CNN → Temporal Aggregation

TABLE IV
FORECASTING AROUSAL EVENTS: PERFORMANCE OF PROPOSED METHOD WITH VARYING PAST DATA LENGTHS ON UNSEEN TEST SET.

	Past data length	90s	180s	270s
Metric				
Accuracy		68.30%	81.31%	67.43%
Sensitivity		53.77%	71.23%	62.97%
Specificity		69.27%	81.90%	67.69%
PPV		9.53%	18.91%	10.23%
NPV		96.13%	97.96%	96.89%
AUROC		0.78	0.88	0.74
AUPRC		0.31	0.39	0.32

TABLE V
FORECASTING AROUSAL EVENTS: PERFORMANCE OF PROPOSED METHOD WITH VARYING PREDICTION HORIZONS ON UNSEEN TEST SET.

	Prediction horizon	30s	60s	120s
Metric				
Accuracy		81.31%	64.01%	62.23%
Sensitivity		71.23%	68.58%	73.94%
Specificity		81.90%	64.59%	61.09%
PPV		18.91%	18.40%	15.63%
NPV		97.96%	94.64%	96.00%
AUROC		0.88	0.77	0.79
AUPRC		0.39	0.33	0.33

→ FC), where the sequential CNN output was temporally aggregated by flattening it into a single vector. Our findings from this experiment indicated that accuracy dropped by approximately 12% and AUPRC by approximately 15%. This proves the critical role of the Bi-LSTM layers in capturing the intricate temporal dependencies within the extracted features for accurate arousal forecasting.

To better understand the contribution of each physiological signal to arousal forecasting, we conducted a signal-wise ablation study. In this experiment, we retrained the model multiple times, each time excluding one input signal, and compared performance against the full model. When all five signals were used, the model achieved an AUPRC of 0.39. The most significant performance drop was observed when EEG was removed, with AUPRC falling to 0.31, highlighting its essential role in capturing cortical signatures of arousals. Removing airflow or SaO2 reduced AUPRC to 0.34 and 0.35, respectively, indicating moderate contributions, likely related to subtle respiratory variability prior to arousals. The exclusion of EMG and EOG resulted in smaller performance declines (AUPRC of 0.36 and 0.37), suggesting they play more supportive roles. These findings are consistent with the physiological relevance of each modality and emphasize the benefit of a multimodal approach, while also pointing to EEG as the most critical component for accurate arousal prediction.

V. DISCUSSION

Accurate prediction of arousals during sleep offers substantial benefits in developing effective preventive measures, therapeutic interventions, and management strategies for sleep-

related issues. In pursuit of this goal, we introduced a novel automated approach that leverages deep learning to model physiological patterns that lead to sleep arousals. Our algorithm employs recurrent convolutional neural networks trained on a diverse array of physiological signals, enabling it to precisely forecast moments of arousal and non-arousal. This predictive capability is crucial for tailoring interventions and improving overall sleep quality, as it provides a deeper understanding of the intricate dynamics that precede sleep disruptions.

To evaluate the efficacy of our forecasting model, we used the PhysioNet Challenge 2018 dataset, a comprehensive compilation of sleep records from 994 individuals with an average age of 55. By adhering to the challenge's guidelines, our focus was on the forecasting of RERA and non-RERA, non-apnea sleep arousals. Our methodology harnessed the potential of deep learning, particularly through the analysis of EEG, EOG, EMG, SaO₂, and AirFlow signal patterns from the past. This strategy enabled us to accurately forecast sleep arousals, demonstrating our model's capability to provide actionable insights for enhancing sleep quality and informing personalized sleep management practices.

Our empirical investigation revealed a critical dependency of model performance on the input sequence length, with an optimal window identified at 180 seconds. Shorter sequences proved insufficient in providing the necessary context for capturing vital temporal dependencies. Conversely, excessively long sequences diluted the relevant 'signal' amidst noise and could exacerbate challenges like vanishing or exploding gradients in certain architectures [40]. Thus, our results underscore the importance of empirically determining the optimal input length to effectively extract meaningful patterns and ensure robust model performance.

Our study primarily focused on predicting future arousals within 30, 60, and 120-second horizons. This choice was driven by the need to align with established clinical practice and standard PSG epoch definitions, ensuring our predictions are directly interpretable and comparable within a clinically relevant framework. While sub-30 second predictions offer intriguing potential for more immediate interventions, exploring these shorter prediction horizons remains a crucial and promising avenue for future research.

Given that our work is the first to focus on forecasting arousals in advance, we have compared the performance of our proposed method with the state-of-the-art arousal detection methods in Table VI. It should be noted that predicting the onset of sleep events is notably more challenging than detecting them post-occurrence. For example, the study by Li et al. [36], published in *Nature Communications Biology*, aimed at detecting both RERA-related and non-RERA, non-apnea arousals, achieved an AUPRC of 0.55. Similarly, Subramanian et al. [41], using the same dataset, focused on the detection of these arousal categories and achieved an AUROC of 0.82 and an AUPRC of 0.24. A recent study presented a semi-supervised hybrid neural network trained on three datasets [44]. It achieved an AUROC of 0.75 and AUPRC of 0.39 in detecting all arousals types. Two recent studies utilized novel deep learning techniques on the MESA dataset—one based on ECG [21] and the other on PPG[23]—and reported

AUPRC scores of 0.62 and 0.63, respectively. In addition to the larger sample size, their inclusion of all arousal types, including apneic events that are typically easier to detect, may have contributed to the higher performance. These efforts in detection, while significant, underscore the complexities in accurately identifying arousals. In contrast, our work, which pioneers in forecasting the future occurrence of sleep arousals, achieves notable results (Accuracy of 81.31%, Sensitivity of 71.23%, Specificity of 81.90%, AUROC of 0.78, and AUPRC of 0.39). This distinction in approach—forecasting versus detection—highlights our significant contribution to the field, emphasizing the necessity of advancing research in predictive sleep event management.

In a comparative analysis against traditional time-series models like MA, AR, and ARIMA, our deep learning model demonstrated superior performance in predicting sleep arousals. Specifically, when tested on the same dataset, MA, AR, and ARIMA models achieved accuracies ranging from 60% to 75%. In contrast, our deep learning model significantly outperformed these methods, achieving an accuracy of 81.31%. This notable improvement can be attributed to the ability of our neural network to effectively capture and integrate the complex interplay of multiple physiological signals (EEG, EOG, EMG, SaO₂, and AirFlow) that precede arousal events. These results underscore the potential of our deep learning approach to advance the field of sleep arousal prediction, offering a more accurate and comprehensive tool for identifying and managing sleep disruptions.

Our results were promising, indicating that the automatic prediction of non-apneic arousals is indeed a feasible task and can be achieved with remarkable accuracy using our algorithm. These findings have significant implications for the diagnosis and treatment of sleep-related disorders, suggesting that our approach has the potential to serve as an effective tool for early detection and management of these conditions. Future work should focus on forecasting both types of arousals to enhance predictive capabilities further. For example, considering that studies like the one by Subramanian et al. [41] have achieved an AUPRC of 0.65 in detecting all arousals types compared to 0.24 for non-apneic arousals, this indicates that the detection of apneic arousals may be more straightforward. Overall, our study represents a significant step forward in the field of sleep research, highlighting the transformative potential of machine learning in this domain.

Although the model's PPV was modest at 18.91%, this result reflects the low prevalence of arousal events in the dataset—a common challenge in sleep monitoring and other rare-event prediction settings. Importantly, the model exhibited high sensitivity and a strong NPV, making it particularly valuable in clinical applications where minimizing false negatives is critical. In such contexts, including real-time alert systems, wearable monitors, or early-warning sleep intervention tools, the priority is often to ensure that potential arousals are not missed—even if that comes at the cost of some false positives. Future work could improve PPV by incorporating post-prediction filtering mechanisms, combining contextual features, or adjusting alert thresholds to balance clinical priorities.

TABLE VI
COMPARISON OF OUR NOVEL AROUSAL FORECASTING METHOD WITH EXISTING AROUSAL DETECTION METHODS (NOTE: THIS STUDY IS THE FIRST TO FORECAST AROUSALS)

Method	Dataset	PSG Signals	Performance				
			ACC	Sen	Spec	AUROC	AUPRC
Feature Extraction + Random Forest (2020) [41]	Physionet-CinC challenge (994 Individuals)	EEG (F3-M2, F4-M1, C3-M2, C4-M1, O1-M2, O2-M1); EOG; EMG (chin, abdominal, chest); ECG	-	-	-	0.82	0.24
Feature Extraction + Feed-forward NN (2020) [42]	Physionet-CinC challenge (994 Individuals)	EEG (F3-M2, F4-M1, C3-M2, C4-M1, O1-M2, O2-M1); EOG; EMG (chin, abdominal, chest); Respiratory airflow; SaO2; ECG	-	0.70	0.70	-	-
CNN + LSTM (2020) [21] [†]	MESA/SHHS (1547/2651 Individuals)	ECG	-	0.65	-	0.93	0.62
Deep U-Net (2021) [36]	Physionet-CinC challenge (994 Individuals)	EEG (F3-M2, F4-M1, C3-M2, C4-M1, O1-M2, O2-M1); EOG; EMG (chin, abdominal, chest); Respiratory airflow; SaO2; ECG	-	-	-	0.93	0.55
Feature Extraction + Meta Classifier (2021) [43]	Physionet-CinC challenge (994 Individuals)	EEG (C3-M2)	0.82	0.84	0.80	0.84	-
Multimodal Attentive (CNN-RNN (2023) [20]) [†]	MrOSSleep Study (2853 Individuals)	EEG (C3-M2, C4-M1); EOG; EMG (Chin, Leg); Nasal pressure; Thoracic & abdominal effort	-	0.61	-	-	-
Hybrid NN with Semi-Supervised Learning (2024) [44] [†]	Physionet/SHHS/Private (100/200/60 Individuals)	EEG (C3-A2, C3-M1, C3-M2)	-	-	-	0.75	0.39
1-D CNN-Based UNET (2024) [22]	Physionet-CinC challenge (994 Individuals)	EEG (F3-M2, F4-M1, C3-M2, C4-M1, O1-M2, O2-M1); EOG; EMG (chin, abdominal, chest); Respiratory airflow; SaO2; ECG	-	-	-	0.94	0.49
Collaborative Learning (2025) [23] [†]	MESA/Private (1896/120 Individuals)	PPG	-	-	-	0.93	0.63
DeepArousal-Net (Proposed Method)	Physionet-CinC challenge (994 Individuals)	EEG (C4-M1); EOG; EMG; Respiratory airflow SaO2	0.81	0.71	0.82	0.88	0.39

[†] Considered all types of arousals.

Envisioning a paradigm where the likelihood of an arousal is anticipated before its occurrence opens the door to innovative intervention strategies, transcending traditional reactive methods. Specifically, beyond the context of apnea-related disturbances already addressed by adaptive CPAP technologies, this predictive capacity invites novel proactive preventive measures for arousals triggered by external factors—such as playing or amplifying white noise in anticipation [45], or real-time modulation of the sleep environment through subtle adjustments to room temperature, light, or dynamic soundscapes [46]. It further enables adaptive therapeutic interventions, by allowing a therapeutic response before an arousal for conditions like chronic pain or restless legs syndrome, potentially through smart bedding systems applying gentle pressure or vibration, or precision-timed delivery of non-pharmacological treatments [47]. Moreover, it supports personalized management strategies, such as alerting caregivers of impending arousal in vulnerable populations to prevent adverse events, providing tailored sleep hygiene advice based on predicted arousal patterns, or enabling more nuanced and preemptive adjustments in adaptive CPAP technologies for non-apnea related arousals. Such approaches open a new era of personalized sleep management, where interventions are not just responsive but anticipatory, tailored to mitigate the risk of arousal with precision and foresight.

During our study, we encountered some limitations, including time and hardware restrictions, which prevented the use of all the available information in the dataset, such as some

EEG channels. Evaluating the proposed method on the entire dataset, using all available physiological and demographic information can further enhance the effectiveness and robustness of our deep learning framework.

Although our study has shown promising results in predicting individual sleep arousal episodes using deep learning techniques, there are still opportunities for further research and improvement. Future research could explore the integration of graph neural networks (GNNs) or transformer-based models, which have shown promise in various domains for their ability to model complex dependencies. GNNs, for instance, could offer a novel approach by treating physiological signals as nodes in a dynamic graph, enabling the capture of non-linear interactions between different signal types. Alternatively, transformer models, renowned for their effectiveness in capturing long-range dependencies in sequence data, could be adapted.

Furthermore, the analysis of the temporal dynamics of sleep arousal episodes is a critical aspect that requires further attention. Investigating the relationships between different sleep stages and arousal events over time could provide important insights into the underlying mechanisms and potentially inform the development of personalized sleep interventions. Finally, exploring the transferability of our method across different populations and sleep disorders could help to establish the generalizability and potential clinical applications of our approach.

In this study, we adopted a comprehensive approach by

leveraging five PSG channels—EEG, EOG, EMG, airflow, and SaO₂—to maximize the model's predictive power. To assess the individual contribution of each modality, we performed an ablation analysis by systematically removing one signal at a time and measuring the corresponding change in AUPRC. This analysis revealed that EEG contributed most significantly to prediction performance, while EOG and EMG had more modest effects. Building on this, future work will explore the integration of explainability techniques such as SHAP (SHapley Additive exPlanations) to further dissect model behavior at the feature level.

Our study demonstrates the effectiveness of deep learning in extracting and analyzing features from various physiological signals, providing strong evidence of its capabilities in predicting the likelihood of arousal occurrence in the future. These results emphasize the potential of this approach as a valuable tool for medical diagnosis and treatment. Our study also lays the groundwork for the development of reliable and accurate algorithms for forecasting other sleep events such as apnea, which can have a significant impact on improving sleep quality and overall health.

VI. CONCLUSIONS

This study proposed DeepArousal-Net, a novel multi-block CNN-LSTM deep learning model for forecasting sleep arousals from multichannel PSG data. The model achieved an accuracy of 81.31%, sensitivity of 71.23%, and specificity of 81.9% in forecasting arousals. We found that the best performance is achieved when using data from 180 seconds prior to the arousal event, and by predicting the occurrence of arousals in the next 30 seconds. Our findings demonstrate the potential of deep learning algorithms in accurately forecasting sleep events such as arousals from multichannel sleep physiological data. This approach can have significant applications in sleep monitoring and management, as well as in wearable devices for sleep enhancement.

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